

Lepton MVA in ONNX

TMVA → ONNX conversion, validation, and what the scores say about our $e\mu$ selection

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What the files turned out to be

Two TMVA weight XMLs, initially assumed to be trigger scale factors.

They are not. No HLT path, no leg p_T/η efficiency map, no data/MC ratio anywhere in them.

They are the **ttH-multilepton lepton MVA** (`mvaTTH`) — a per-lepton prompt-vs-nonprompt discriminant.

Creator / date	<code>cvico</code> , 2023-12-20
Origin	<code>CMGTools/TTHAnalysis</code> , CMSSW_12_4_12, wz-run3
Method	<code>BDT::BDTG</code> — 500 trees, MaxDepth 8, shrinkage 0.1
Classes	binary: Signal = prompt, Background = nonprompt
Transforms	<code>NTransformations="0"</code> — no input preprocessing
Training events	879,894 (muon)

Why not just read the NanoAOD branch?

Muon_mvaTTH and Electron_mvaTTH **do exist** in NanoAODv12 — verified on a central TTto2L2Nu v12 file.

We do not use them:

- the stock branch is a **Run 2** training
- these XMLs are a dedicated **2022EE** retraining
- same 13 variables, **different weights**

The stock branch would be **subtly** wrong, not obviously wrong — the worst kind. Evaluate these weights instead.

All 13 input branches verified present for **both** flavours in NanoAODv12 — 0 missing.

The 13 inputs

i	muon	electron
0–1	pt, eta	pt, eta
2–4	pfRelIso03_all, miniPFRelIso_chg, neutral remainder	same
5–6	jetNDauCharged, jetPtRelv2	same
7	jetIdx>-1 ? Jet_btagDeepFlavB[jetIdx] : 0	same
8	min(1/(1+jetRelIso), 1.5)	same
9–11	sip3d, `log	dxy
12	segmentComp	mvaIso

Isolation + jet association + impact parameters + one detector-specific variable.

Slot 7 is a **conditional cross-collection lookup**, not a plain branch.

The response formula

TMVA BoostType=Grad, two classes:

```
raw    = sum_t res_t(x)           # leaf 'res' values summed, no extra factor
score = 2 / (1 + exp(-2*raw)) - 1 # in (-1, 1)
```

Shrinkage is already folded into the stored leaf values. Do not reapply it.

Verified empirically rather than assumed:

	leaf <code>res</code> absmax
tree 0	0.0999 $\approx$ shrinkage 0.1
tree 499	0.0224

The cap at ~ 0.1 and the decay across the ensemble are exactly what gradient boosting with pre-folded shrinkage produces.

Mapping TMVA trees onto ONNX

Node attributes in the XML:

attribute	meaning
IVar	feature index (-1 on leaves)
Cut	split threshold
nType	-99 marks a leaf
pos	s / l / r — root, left, right
res	leaf response

TMVA sends `var <= Cut` to the **left** child.

ONNX `BRANCH_LEQ` has the same convention → `true_branch = left`.

Emitted as a `TreeEnsembleRegressor` (raw sum) followed by the closed-form squashing, so the graph reproduces the **full score**, not just the raw sum.

Validation: three implementations

1. ONNX

`TreeEnsembleRegressor` + squashing

2. numpy

independent recursive mask-based tree walk

3. `TMVA::Reader`

the reference implementation, via LCG view

`LCG_104a/x86_64-el9-gcc12-opt`

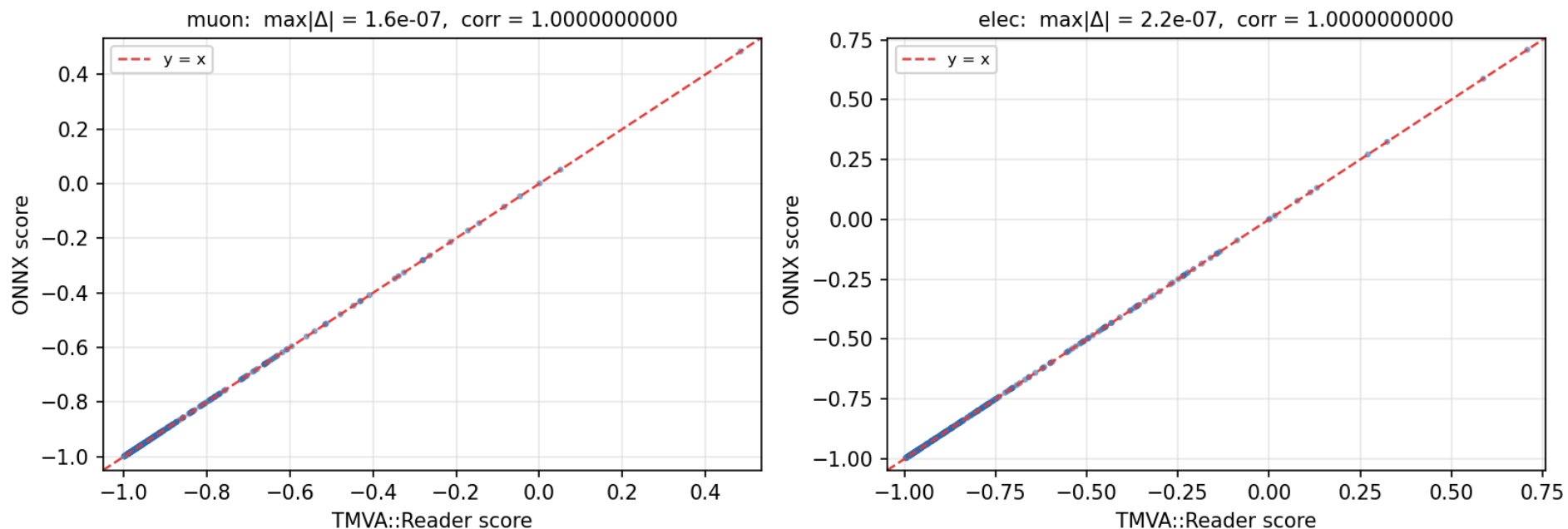
`TMVA::Reader` is slow (scalar API, 500 trees): 2000 points times out at 2 min;
300 points suffices.

ONNX-vs-numpy alone is **not sufficient** —
both encode *my* reading of the
format.

Only `TMVA::Reader` can catch an inverted
branch or a double-applied
shrinkage.

Validation result

ONNX vs TMVA::Reader — identical to float32 round-off



pair	N	max $ \Delta $	corr
ONNX vs numpy (muon / elec)	2000	2.2e-07 / 2.9e-07	—
ONNX vs TMVA::Reader (muon)	300	1.6e-07	1.0000000000
ONNX vs TMVA::Reader (elec)	300	2.2e-07	1.0000000000

So: should we use it?

The conversion succeeded. That does not by itself mean the cut belongs in our analysis.

The deciding question:

How much nonprompt background is actually in our selection?

Measured directly — central `TTto2L2Nu` v12, 400k events, with the `hww_combine_2dcat` lepton selection reproduced:

- muon: tight ID + tight PF iso, $p_T > 10$, $|\eta| < 2.4$
- electron: wp80iso, $p_T > 10$, $|\eta| < 2.5$
- exactly one of each, opposite sign, lead $p_T > 20$, sublead > 10

49,070 $e\mu$ events survive. Truth label from `genPartFlav`.

The headline number

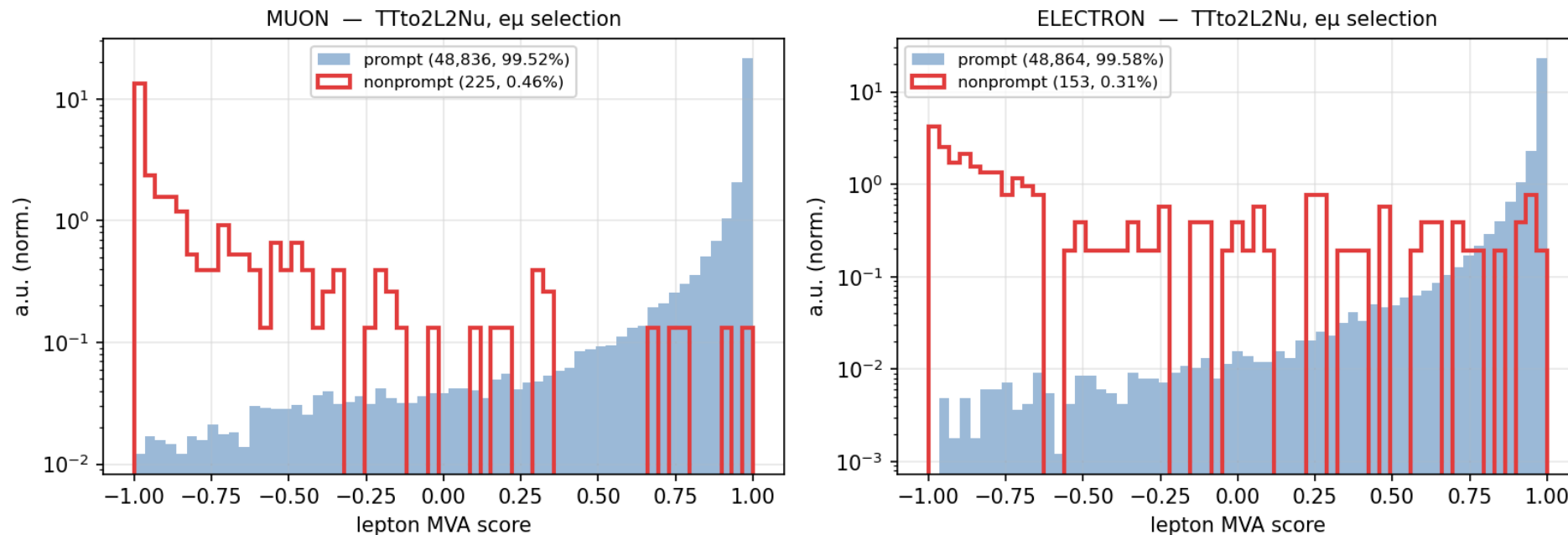
flavour	prompt	nonprompt	other
muon	48,836 — 99.52%	225 — 0.46%	9 — 0.02%
electron	48,864 — 99.58%	153 — 0.31%	53 — 0.11%

Nonprompt leptons are 0.3–0.5% of the selection. The existing cut-based tight ID + isolation has already removed essentially all of them.

The MVA is aimed at a background we have already eliminated by other means.

The separation itself is real

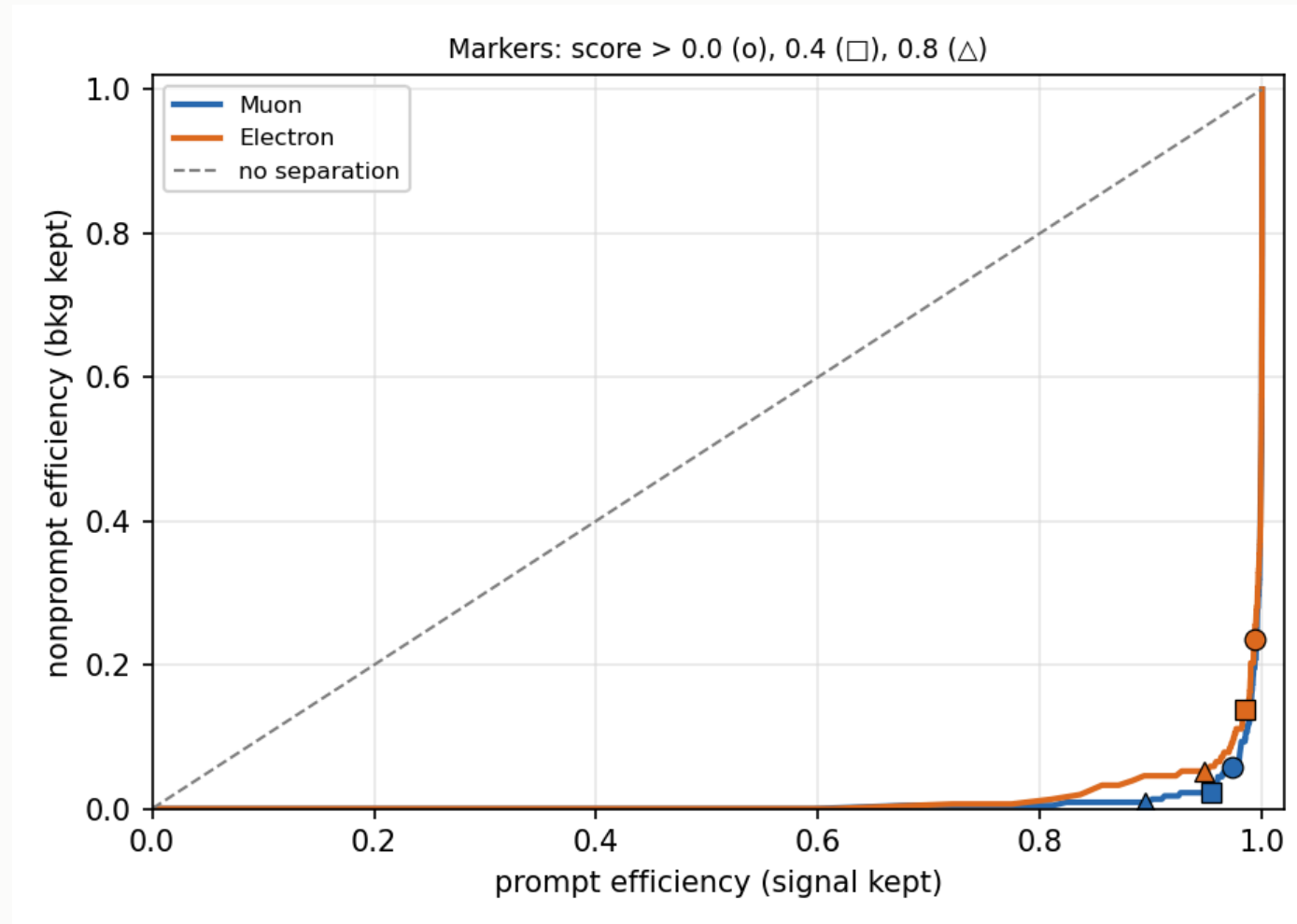
Separation is real — but nonprompt is only 0.3–0.5% of the selection



	muon	electron
median score, prompt	+0.9952	+0.9945
median score, nonprompt	-0.9510	-0.7640

Electron separation is weaker — `mvaIso` sits in slot 12 and already overlaps heavily with the `wp80iso` cut applied upstream.

What a cut would cost



Muon working points:

cut	eff(prompt)	eff(nonprompt)	rejection
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The trade, quantified

Taking `score > 0.4` on both legs:

removes	~98% of an already-0.46% contamination → ~0.45 pp
costs	~4.5% of prompt muons + ~1.5% of prompt electrons → ~6% of signal

~6% of signal efficiency to remove ~0.45% of background.

The loss lands directly on H+c signal — whose limit is already dominated by MC statistics (autoMCStats = -255 of the -523 total systematic budget).

No reference analysis of ours does this

analysis	lepton treatment
AN-23-102 (our Run 2 predecessor)	no lepton MVA
HH $\rightarrow$ bbWW Run 3 (same $e\mu$, 13.6 TeV)	cut-based ID + PF rel-iso
ttH-multilepton (2011.03652)	source of these weights — "tight lepton" = loose + MVA cut

ttH-multilepton works in **same-sign 2 ℓ / 3 ℓ / 4 ℓ** , where nonprompt is a *leading* background and there is no OS continuum.

Ours is **OS $e\mu$, 82% real $t\bar{t}$** , both leptons genuinely prompt.

The technique is imported from a different topology — the 0.4% measurement is the quantitative version of that observation.

The calibration cost, if we did adopt it

From the ttH paper's efficiency section — once "tight" is defined by an MVA cut:

1. measure efficiency by **tag-and-probe in $Z \rightarrow \ell\ell$**
2. cross-check in an **OS $e\mu + \geq 2$ jets $t\bar{t}$ CR**, nonprompt subtracted via an SS $e\mu$ sideband
3. take the DY-vs- $t\bar{t}$ difference as a **1–2% systematic**

That $t\bar{t}$ control region **is our signal region**.

And SFs for these specific 2022EE weights are unlikely to exist publicly.

Plus a full reprocessing campaign.

Recommendation

Do not adopt a lepton-MVA cut in the $e\mu$ signal region. The cost/benefit is clearly negative on these numbers.

Where the converted models could still earn their keep:

use	note
Nonprompt/QCD control region	<i>invert</i> the MVA → clean enriched region. Much better use than cutting in the SR.
Input feature, not a cut	costs no signal efficiency by construction; needs retraining, and gain is small at 0.4%
Evidence the selection is already clean	this measurement itself — a publishable statement

Caveats

- **Single TT file**, 400k events → 225 nonprompt muons, 153 electrons.
Adequate for a 0.4%-vs-6% decision; too coarse for a precise WP scan.
- **$t\bar{t}$ only**. V+jets/QCD have higher nonprompt fractions, but they are 7.3% and ~0% of the SR — the conclusion is not sensitive to this.
- **$\log|d_{xy}|$ floor unresolved**. Used $1e-6$; the training producer's convention is unknown. Affects few leptons, but must be pinned before any production use.

Artifacts — all durable on EOS

```
/eos/user/c/cgupta/HToWW/leptonmva/  
onnx/muon_mvaTTH_2022EE.onnx      0.38 MB  <- validated  
onnx/electron_mvaTTH_2022EE.onnx  0.40 MB  <- validated  
tmva_to_onnx.py                   converter (checks Options, refuses transforms)  
numpy_ref.py                      independent numpy evaluator  
tmva_reference.py                 TMVA::Reader reference (needs LCG view)  
validate_vs_tmva.py              the comparison  
tmva_reference.npz               frozen reference inputs + scores  
diagnostic_nonprompt.py          the ep measurement
```

Re-run validation at any time:

```
cd /eos/user/c/cgupta/HToWW/leptonmva  
$MAMBA_EXE run -n b_hive python3 validate_vs_tmva.py
```

No production code was modified.

Summary

Conversion: exact — corr = 1.0000000000 vs TMVA::Reader

Nonprompt in our $e\mu$ selection: 0.3–0.5%

A cut would cost ~6% of signal to remove ~0.45% of background

Recommend: do not cut. Keep the models for a nonprompt CR.